EXP 801 DS Wheat Heat Drought

— Ed Chaplin (USyd), *Apr–June 2025* —

Analysis of anatomical traits

Load packages

```
library(knitr)
#knitr::spin("Exp801.anat.anal.v2.r")
library(ggplot2)
library(asreml)
library(asremlPlus)
library(parallel) #for parallel processing
library(foreach)
library(doParallel)
packageVersion("asreml")
```

```
[1] '4.2.0.481'
```

```
packageVersion("asremlPlus")
```

```
[1] '4.4.58'
```

Set up constants and files

```
options(width=95)
source("Exp801.anat.anal.v1.stub.r")
load("../data/anat.dat.v2.Rdata")
saveFile <- "../data/anat.Analyses.v2.RData"
```

Initialize parallel processing

```
ncores <- detectCores()
cl <- makeCluster(ncores)
registerDoParallel(cl)

responses.pot <- responses.anat[c(1, 2, 7)]
responses.stoma <- responses.anat[!(responses.anat %in% responses.pot)]

# Fit the models to the pot level first
```

```
kvec <- 1:length(responses.anat)
models <- foreach(response = responses.anat[kvec],
                  .errorhandling = "pass",
                  .inorder=TRUE,
                  .packages = c("asreml", "asremlPlus", "dae"))  %dopar%
  {
    if(response %in% responses.pot){
      mod <- fitIndvModel(response=response, selection = "backward",
                          fix.mod = ~ ASide/ABlock + Rm_Temperature/(AWatering*ALine + HWatering*HLine)
                          ran.mod = ~  Room:Side/Location/SubUnit,
                          term.marginality = marginality(mod.canon)$trts,
                          classify = c("Rm_Temperature:AWatering:ALine:HWatering:HLine"),
                          add.facs2preds = c("Watering", "Line", "Line.ID", "WT"),
```

```
                          hetero.facs =  c("Watering", "Rm_Temperature"),
                          dat = anat.pot.dat)
    } else{
      mod <- fitIndvModel(response=response, selection = "backward",
                          fix.mod = ~ ASide/ABlock + Rm_Temperature/(AWatering*ALine + HWatering*HLine),
                          ran.mod = ~  Room:Side/Location/SubUnit/Cell,
                          term.marginality = marginality(mod.canon)$trts,
                          classify = c("Rm_Temperature:AWatering:ALine:HWatering:HLine"),
                          add.facs2preds = c("Watering", "Line", "Line.ID", "WT"),
                          hetero.facs =  c("Watering", "Rm_Temperature"),
                          dat = anat.stoma.dat)
    }
  }
names(models) <- responses.anat[kvec]

msgs <- (do.call(rbind, lapply(models,
                               function(mod) if ("message" %in% names(mod)) return(mod$message))))
if (!is.null(msgs))
{
  colnames(msgs) <- "messages"
  cat("\n#### Output any error messages\n\n")
  print(msgs)
}
```

Hand back clusters

```
stopCluster(cl)
```

Save workspace - the compression takes longer, but produces a substantially smaller file

```
save.image(saveFile, compress = "xz")
```